

ALFIDO TECH INTERNSHIP

Submitted By: Palak Vij

Internship Domain: C / C++ Programming

Internship Duration: 10 April 2026 – 10 May 2026

Tools Used: GCC, G++, VS Code, GitHub, GDB

This internship project focused on strengthening programming fundamentals in C and C++ through practical implementation of data structures, algorithms, and system programming concepts. The tasks emphasized clean coding practices, logical problem solving, memory management, debugging, and real-world execution using professional development tools.

TASK 1: DATA STRUCTURES IMPLEMENTATION (C++)

Implemented core data structures including Linked List and Stack using C++. The project demonstrated insertion, deletion, traversal, push/pop operations, and memory handling. The objective of this task was to understand the internal working of dynamic data structures and improve logical implementation skills. **Key Features:**

- Linked List insertion and deletion operations
- Stack implementation using arrays
- Display functionality for visualization
- Complexity awareness and memory handling

Compile & Run Commands:

```
g++ main.cpp -o output
./output
```

TASK 1: DATA STRUCTURES (C++)

Program:

Linked List and Stack

Description:

Implemented Linked List operations (insert, delete, display) and Stack operations (push, pop, display).

```
C:\Windows\System32\cmd.exe

--- LINKED LIST OPERATIONS ---
Linked List after insertions:
30 -> 20 -> 10 -> NULL
After deleting 20:
30 -> 10 -> NULL

--- STACK OPERATIONS ---
Stack after push operations:
3 2 1
After one pop operation:
2 1
Program executed successfully.
```

TASK 2: ALGORITHMS (C++)

Program:

Merge Sort, Binary Search and Activity Selection (Greedy)

Description:

Implemented Merge Sort for sorting, Binary Search for searching an element, and Activity Selection using Greedy approach.

```
C:\Windows\System32\cmd.exe

--- MERGE SORT ---
Sorted Array:
1 2 3 5 9

--- BINARY SEARCH ---
Element to search: 3
Binary Search: Found 3 at index 2

--- ACTIVITY SELECTION (GREEDY) ---
Selected Activities (Start, Finish):
(1, 2) (3, 4) (5, 7) (8, 9)
Program executed successfully.
```

TASK 3: SYSTEM PROGRAMMING (C)

Program:

Producer-Consumer using Pipes

Description:

Implemented inter-process communication between Producer and Consumer using pipes in C.

```
C:\Windows\System32\cmd.exe

--- PRODUCER-CONSUMER USING PIPES ---
Produced: 100
Consumed: 100

Program executed successfully.
```

TASK 2: ALGORITHMS & PROBLEM SOLVING (C++)

Implemented important algorithmic techniques such as Merge Sort, Binary Search, and Greedy Activity Selection. This task focused on optimizing problem solving approaches and understanding time complexity. **Key Features:**

- Merge Sort using Divide & Conquer
- Binary Search for efficient searching
- Activity Selection using Greedy approach
- Runtime and complexity understanding

Compile & Run Commands:

```
g++ main.cpp -o output
```

```
./output
```

TASK 1: DATA STRUCTURES (C++)

Program:

Linked List and Stack

Description:

Implemented Linked List operations (insert, delete, display) and Stack operations (push, pop, display).

```
C:\Windows\System32\cmd.exe

--- LINKED LIST OPERATIONS ---
Linked List after insertions:
30 -> 20 -> 10 -> NULL
After deleting 20:
30 -> 10 -> NULL

--- STACK OPERATIONS ---
Stack after push operations:
3 2 1
After one pop operation:
2 1
Program executed successfully.
```

TASK 2: ALGORITHMS (C++)

Program:

Merge Sort, Binary Search and Activity Selection (Greedy)

Description:

Implemented Merge Sort for sorting, Binary Search for searching an element, and Activity Selection using Greedy approach.

```
C:\Windows\System32\cmd.exe

--- MERGE SORT ---
Sorted Array:
1 2 3 5 9

--- BINARY SEARCH ---
Element to search: 3
Binary Search: Found 3 at index 2

--- ACTIVITY SELECTION (GREEDY) ---
Selected Activities (Start, Finish):
(1, 2) (3, 4) (5, 7) (8, 9)
Program executed successfully.
```

TASK 3: SYSTEM PROGRAMMING (C)

Program:

Producer-Consumer using Pipes

Description:

Implemented inter-process communication between Producer and Consumer using pipes in C.

```
C:\Windows\System32\cmd.exe

--- PRODUCER-CONSUMER USING PIPES ---
Produced: 100
Consumed: 100

Program executed successfully.
```

TASK 3: SYSTEM PROGRAMMING (C)

Developed a Producer-Consumer communication program using Pipes and Process Creation in C. This task introduced system-level programming concepts including inter-process communication and UNIX system calls. **Key Features:**

- Process creation using `fork()`
- Pipe-based communication
- Producer-Consumer implementation
- Error handling and process synchronization

Compile & Run Commands:

```
gcc main.c -o output
```

```
./output
```

TASK 1: DATA STRUCTURES (C++)

Program:

Linked List and Stack

Description:

Implemented Linked List operations (insert, delete, display) and Stack operations (push, pop, display).

```
C:\Windows\System32\cmd.exe
--- LINKED LIST OPERATIONS ---
Linked List after insertions:
30 -> 20 -> 10 -> NULL
After deleting 20:
30 -> 10 -> NULL

--- STACK OPERATIONS ---
Stack after push operations:
3 2 1
After one pop operation:
2 1
Program executed successfully.
```

TASK 2: ALGORITHMS (C++)

Program:

Merge Sort, Binary Search and Activity Selection (Greedy)

Description:

Implemented Merge Sort for sorting, Binary Search for searching an element, and Activity Selection using Greedy approach.

```
C:\Windows\System32\cmd.exe
--- MERGE SORT ---
Sorted Array:
1 2 3 5 9

--- BINARY SEARCH ---
Element to search: 3
Binary Search: Found 3 at index 2

--- ACTIVITY SELECTION (GREEDY) ---
Selected Activities (Start, Finish):
(1, 2) (3, 4) (5, 7) (8, 9)
Program executed successfully.
```

TASK 3: SYSTEM PROGRAMMING (C)

Program:

Producer-Consumer using Pipes

Description:

Implemented inter-process communication between Producer and Consumer using pipes in C.

```
C:\Windows\System32\cmd.exe
--- PRODUCER-CONSUMER USING PIPES ---
Produced: 100
Consumed: 100

Program executed successfully.
```

CONCLUSION

The Alfido Tech Internship provided practical exposure to real-world C and C++ programming concepts. The tasks improved logical thinking, debugging capabilities, algorithmic understanding, and system-level programming knowledge. The project also enhanced software development practices such as documentation, GitHub management, and structured coding methodologies.